MICROSERVICES / EVENT-DRIVEN ARCHITECTURE

DECOUPLE TO SURVIVE

Tight coupling is the root cause of cascading failures. When services share state or deploy together, one failure becomes many. Loose coupling — through message queues, circuit breakers, and independent deployment — means failures stay contained.

KEY BENEFITS

- ✓ Failures are isolated — one component failing cannot take the whole system down
- ✓ Services can be deployed, scaled, and updated independently
- ✓ Teams can work and release in parallel without coordination bottlenecks
- ✓ Enables incremental modernisation — replace one service without touching others

REAL-WORLD EXAMPLE

Amazon's 2002 Bezos API mandate — all teams expose capabilities through service interfaces — created the architectural foundation for AWS. The decoupling enforced by that decision became a $90B cloud business.

PRACTICAL EXAMPLE

The email service goes down. Because order confirmations are sent via an async queue, checkout continues working. When email recovers it drains the backlog. Customers get emails 45 minutes late. No orders are lost.

REFLECTION QUESTION

If your most critical downstream service went down right now, how many user-facing features would fail with it?